Sharif University of Technology
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Coding the `Trans` and `Rot` Functions and Using Them for Coordinate Frame Transformations

Robotics Course – Homework 1

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1 Problem Statement

1.1 Question 1

Given the MATLAB code provided for the homogeneous translation transform (`Trans`), implement similar code for the homogeneous rotation transform in MATLAB.

The MATLAB function given in the problem statement for the homogeneous translation transform:

```
function T = Trans(axis, distance)
%Homogenous transformation along "axis" for the amount of "distance"

axis = upper(axis);
switch axis
    case 'X'
        T = [1 0 0 distance; 0 1 0 0; 0 0 1 0; 0 0 0 1];
    case 'Y'
        T = [1 0 0 0; 0 1 0 distance; 0 0 1 0 ; 0 0 0 1];
    case 'Z'
        T = [1 0 0 0; 0 1 0 0; 0 0 1 distance; 0 0 0 1];
end
```

Note: for the sake of code efficiency, a small modification was made – instead of three `if` statements, a single `switch` statement is used to write this function.

1.2 Question 2

In the second part, given the image in the problem statement, find the position of the blue cube relative to frame 0 using the product of homogeneous transformations.

[Figure: a wedge-shaped solid with edges a, b, c, d, e; frame 0 (X_0, Y_0, Z_0) at one corner of the base and frame 1 (X_1, Y_1, Z_1) at the blue cube, on the top face.]

2 Solutions

2.1 Answer to Question 1

To write the rotation transform code, we first write the standard homogeneous rotation transforms:

$$H_{x,\alpha} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha & 0 \\ 0 & \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_{y,\beta} = \begin{bmatrix} \cos\beta & 0 & \sin\beta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\beta & 0 & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_{z,\gamma} = \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 & 0 \\ \sin\gamma & \cos\gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

We now implement these matrices in MATLAB as a single function:

```
function R = Rot(axis, angle)
%Homogenous rotation about "axis" for "angle" degrees

axis = upper(axis);
switch axis
    case 'X'
        R = [1 0 0 0; 0 cosd(angle) -sind(angle) 0; 0 sind(angle) cosd(
            angle) 0; 0 0 0 1];
    case 'Y'
        R = [cosd(angle) 0 sind(angle) 0; 0 1 0 0; -sind(angle) 0 cosd(
            angle) 0 ; 0 0 0 1];
    case 'Z'
        R = [cosd(angle) -sind(angle) 0 0; sind(angle) cosd(angle) 0 0; 0
            0 1 0; 0 0 0 1];
end
```

It is worth noting that, as in the first part, a `switch` statement is used to make the code more efficient. Also, the function's `angle` input is taken in degrees.

2.2 Answer to Question 2

We implement the answer to this question as a MATLAB script.

The dimensions given in the figure are:

$$a = 20\,\text{cm}, \quad b = 20\,\text{cm}, \quad c = 40\,\text{cm}, \quad d = 15\,\text{cm}, \quad e = 15\,\text{cm}$$

It is clear that transforming frame 0 into frame 1 requires three translations and one rotation. The translations are, in order: along Y by $(a+b) = 40$ cm, then along Z by $(e+5) = 20$ cm, and finally along X by 5 cm. We then apply the rotation R_y about the Y axis by $-\frac{\pi}{2}$, i.e. $-90°$.

We implement the steps above as follows:

$$H_1^0 = \text{Trans}(Y, 40) \cdot \text{Trans}(Z, 20) \cdot \text{Trans}(X, 5) \cdot \text{Rot}(Y, -90)$$

We take the position of the blue cube to be the origin of the resulting coordinate frame (frame 1), written in homogeneous form:

$$O_1^1 = [0, 0, 0, 1]^T$$

To obtain this position in frame 0, we compute:

$$O_1^0 = H_1^0\, O_1^1$$

The MATLAB script written to obtain H_1^0 and compute O_1^0 is as follows:

```
clc
clear

O11 = [0,0,0,1]'; %Homogenous representation of O1 in Frame 1
a = 20;
b = 20;
c = 40;
d = 15;
e = 15;
```

```
%Homogenous transformation from Frame 0 to Frame 1 using matrix
%multiplication
H01 = Trans('Y',a+b)*Trans('Z',e+5)*Trans('X',5)*Rot('Y',-90);

O01 = H01*O11 %Homogenous representation of O1 in Frame 0
```

*The code and related files are included in the appendix (`code/`).